

<h2 id="ANALYSIS-PYTHON-PROJECT---SUPERSTORE-SALES-DATA">ANALYSIS PYTHON PROJECT - SUPERSTORE SALES DATA</h2>

<h2 id="IMPORT-LIBRARIES">IMPORT LIBRARIES</h2>

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

<h2 id="OBJECTIVES">OBJECTIVES</h2>

```
In [ ]: To analyse Superstore sales data and identify sales trends, profit performance, and business insight using python
```

<h2 id="LOAD-DATASET">LOAD DATASET</h2>

```
In [2]: df = pd.read_csv(r"C:\Users\HP\Documents\sample superstore.csv",encoding="latin1")
print(df)
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	\
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	
...	...	...	...	...	...	
9989	9990	CA-2014-110422	1/21/2014	1/23/2014	Second Class	
9990	9991	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	
9991	9992	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	
9992	9993	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	
9993	9994	CA-2017-119914	5/4/2017	5/9/2017	Second Class	

	Customer ID	Customer Name	Segment	Country	City	\
0	CG-12520	Claire Gute	Consumer	United States	Henderson	
1	CG-12520	Claire Gute	Consumer	United States	Henderson	
2	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	
3	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	
4	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	
...	...	...	...	...	...	
9989	TB-21400	Tom Boeckenhauer	Consumer	United States	Miami	
9990	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	
9991	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	
9992	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	
9993	CC-12220	Chris Cortes	Consumer	United States	Westminster	

	...	Postal Code	Region	Product ID	Category	Sub-Category	\
0	...	42420	South	FUR-BO-10001798	Furniture	Bookcases	
1	...	42420	South	FUR-CH-10000454	Furniture	Chairs	
2	...	90036	West	OFF-LA-10000240	Office Supplies	Labels	
3	...	33311	South	FUR-TA-10000577	Furniture	Tables	
4	...	33311	South	OFF-ST-10000760	Office Supplies	Storage	
...	...	...	...	...	...	...	
9989	...	33180	South	FUR-FU-10001889	Furniture	Furnishings	
9990	...	92627	West	FUR-FU-10000747	Furniture	Furnishings	
9991	...	92627	West	TEC-PH-10003645	Technology	Phones	
9992	...	92627	West	OFF-PA-10004041	Office Supplies	Paper	
9993	...	92683	West	OFF-AP-10002684	Office Supplies	Appliances	

		Product Name	Sales	Quantity	\
0		Bush Somerset Collection Bookcase	261.9600	2	
1		Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	
2		Self-Adhesive Address Labels for Typewriters b...	14.6200	2	
3		Bretford CR4500 Series Slim Rectangular Table	957.5775	5	
4		Eldon Fold 'N Roll Cart System	22.3680	2	
...		...	...	...	
9989		Ultra Door Pull Handle	25.2480	3	
9990		Tenex B1-RE Series Chair Mats for Low Pile Car...	91.9600	2	
9991		Aastra 57i VoIP phone	258.5760	2	
9992		It's Hot Message Books with Stickers, 2 3/4" x 5"	29.6000	4	
9993		Acco 7-Outlet Masterpiece Power Center, Wihtou...	243.1600	2	

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714
3	0.45	-383.0310
4	0.20	2.5164
...	...	...
9989	0.20	4.1028
9990	0.00	15.6332
9991	0.20	19.3932
9992	0.00	13.3200
9993	0.00	72.9480

[9994 rows x 21 columns]

<h2 id="DATA-UNDERSTANDING">DATA UNDERSTANDING</h2>

In [3]: df.head()

Out[3]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Row ID</th> <th>Order ID</th> <th>Order Date</th> <th>Ship Date</th> <th>Ship Mode</th> <th>Customer ID</th> <th>Customer Name</th> <th>Segment</th> <th>Country</th> <th>City</th> <th>...</th> <th>Postal Code</th> <th>Region</th> <th>Product ID</th> <th>Category</th> <th>Sub-Category</th> <th>Product Name</th> <th>Sales</th> <th>Quantity</th> <th>Discount</th> <th>Profit</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> </tbody></div><table><tbody><tr><td>1</td><td>CA-2016-152156</td><td>11/8/2016</td><td>11/11/2016</td><td>Second Class</td><td>CG-12520</td><td>Claire Gute</td><td>Consumer</td><td>United States</td><td>Henderson</td><td>...</td><td>42420</td><td>South</td><td>FUR-BO-10001798</td><td>Furniture</td><td>Bookcases</td><td>Bush Somerset Collection Bookcase</td><td>261.9600</td><td>2</td><td>0.00</td><td>41.9136</td></tr><tr><td>2</td><td>CA-2016-152156</td><td>11/8/2016</td><td>11/11/2016</td><td>Second Class</td><td>CG-12520</td><td>Claire Gute</td><td>Consumer</td><td>United States</td><td>Henderson</td><td>...</td><td>42420</td><td>South</td><td>FUR-CH-10000454</td><td>Furniture</td><td>Chairs</td><td>Hon Deluxe Fabric Upholstered Stacking Chairs,...</td><td>731.9400</td><td>3</td><td>0.00</td><td>219.5820</td></tr><tr><td>3</td><td>CA-2016-138688</td><td>6/12/2016</td><td>6/16/2016</td><td>Second Class</td><td>DV-13045</td><td>Darrin Van Huff</td><td>Corporate</td><td>United States</td><td>Los Angeles</td><td>...</td><td>90036</td><td>West</td><td>OFF-LA-10000240</td><td>Office Supplies</td><td>Labels</td><td>Self-Adhesive Address Labels for Typewriters b...</td><td>14.6200</td><td>2</td><td>0.00</td><td>6.8714</td></tr><tr><td>4</td><td>US-2015-108966</td><td>10/11/2015</td><td>10/18/2015</td><td>Standard Class</td><td>SO-20335</td><td>Sean O'Donnell</td><td>Consumer</td><td>United States</td><td>Fort Lauderdale</td><td>...</td><td>33311</td><td>South</td><td>FUR-TA-10000577</td><td>Furniture</td><td>Tables</td><td>Bretford CR4500 Series Slim Rectangular Table</td><td>957.5775</td><td>5</td><td>0.45</td><td>-383.0310</td></tr><tr><td>5</td><td>US-2015-108966</td><td>10/11/2015</td><td>10/18/2015</td><td>Standard Class</td><td>SO-20335</td><td>Sean O'Donnell</td><td>Consumer</td><td>United States</td><td>Fort Lauderdale</td><td>...</td><td>33311</td><td>South</td><td>OFF-ST-10000760</td><td>Office Supplies</td><td>Storage</td><td>Eldon Fold 'N Roll Cart System</td><td>22.3680</td><td>2</td><td>0.20</td><td>2.5164</td></tr></tbody></table><p>5 rows × 21 columns</p>

In [15]: df.describe()

Out[15]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Row ID</th> <th>Postal Code</th> <th>Sales</th> <th>Quantity</th> <th>Discount</th> <th>Profit</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> </tbody></div><table><tbody><tr><td>9994.000000</td><td>9994.000000</td><td>9994.000000</td><td>9994.000000</td><td>9994.000000</td><td>9994.000000</td></tr><tr><td>4997.500000</td><td>55190.379428</td><td>229.858001</td><td>3.789574</td><td>0.156203</td><td>28.656896</td></tr><tr><td>2885.163629</td><td>32063.693350</td><td>623.245101</td><td>2.225110</td><td>0.206452</td><td>234.260108</td></tr><tr><td>1.000000</td><td>1040.000000</td><td>0.444000</td><td>1.000000</td><td>0.000000</td><td>-6599.978000</td></tr><tr><td>2499.250000</td><td>23223.000000</td><td>17.280000</td><td>2.000000</td><td>0.000000</td><td>1.728750</td></tr><tr><td>4997.500000</td><td>56430.500000</td><td>54.490000</td><td>3.000000</td><td>0.200000</td><td>8.666500</td></tr><tr><td>7495.750000</td><td>90008.000000</td><td>209.940000</td><td>5.000000</td><td>0.200000</td><td>29.364000</td></tr><tr><td>9994.000000</td><td>99301.000000</td><td>22638.480000</td><td>14.000000</td><td>0.800000</td><td>8399.976000</td></tr></tbody></table>

In [16]: df.info()

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Row ID          9994 non-null   int64
1   Order ID        9994 non-null   str
2   Order Date      9994 non-null   str
3   Ship Date       9994 non-null   str
4   Ship Mode       9994 non-null   str
5   Customer ID     9994 non-null   str
6   Customer Name   9994 non-null   str
7   Segment        9994 non-null   str
8   Country         9994 non-null   str
9   City            9994 non-null   str
10  State           9994 non-null   str
11  Postal Code     9994 non-null   int64
12  Region          9994 non-null   str
13  Product ID      9994 non-null   str
14  Category        9994 non-null   str
15  Sub-Category    9994 non-null   str
16  Product Name    9994 non-null   str
17  Sales           9994 non-null   float64
18  Quantity        9994 non-null   int64
19  Discount        9994 non-null   float64
20  Profit          9994 non-null   float64
dtypes: float64(3), int64(3), str(15)
memory usage: 1.6 MB
```

In []: The dataset contains sales,profit customer,product,and regional information. The data is well-structured and suitable for sales analysis

```
<h2 id="DATA-CLEANING">DATA CLEANING</h2>
```

```
<h2 id="FIND-NULL-VALUE">FIND NULL VALUE</h2>
```

```
In [17]: df.isnull().sum()
```

```
Out[17]: Row ID          0
Order ID          0
Order Date        0
Ship Date         0
Ship Mode         0
Customer ID       0
Customer Name     0
Segment          0
Country           0
City              0
State             0
Postal Code       0
Region            0
Product ID        0
Category          0
Sub-Category      0
Product Name      0
Sales             0
Quantity          0
Discount          0
Profit            0
dtype: int64
```

In []: No,missing values were found in the dataset, indicating that the data is complete and ready for analysis.

```
<h2 id="FIND-DUPLICATE-VALUE">FIND DUPLICATE VALUE</h2>
```

```
In [19]: df.duplicated().sum()
```

Out[19]: np.int64(0)

In []: No,duplicate records were found,ensuring accurate and reliable analysis.

<h2 id="FIND-SUM-OF-SALES">FIND SUM OF SALES</h2>

In [21]: df["Sales"].sum()

Out[21]: np.float64(2297200.8603)

<h2 id="FIND-SUM-OF-PROFIT">FIND SUM OF PROFIT</h2>

In [24]: df["Profit"].sum()

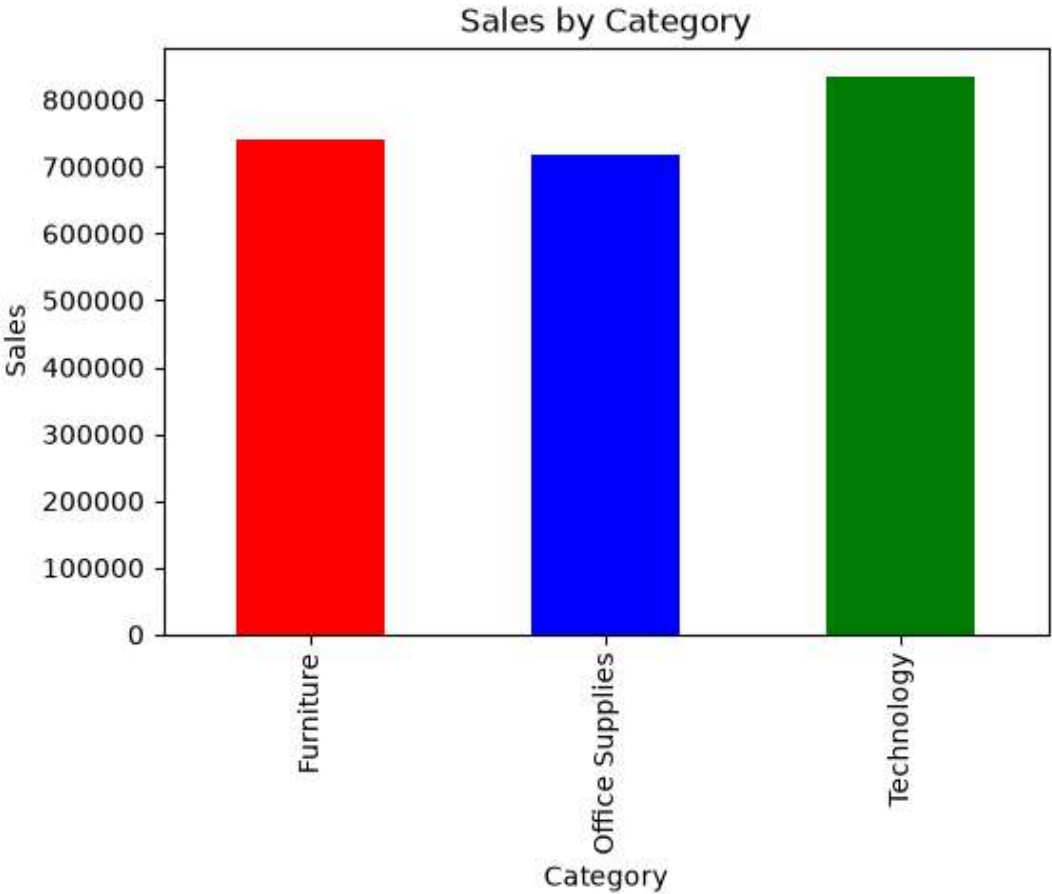
Out[24]: np.float64(286397.0217)

In []: The business generated a total sales value of approximately 2.30 million, with an overall profit of approximately 286 thousand, indicating positive busiess performance.

<h2 id="BAR-CHART">BAR CHART</h2><h2 id="SALES-BY-CATEGORY">SALES BY CATEGORY</h2>

In [5]: category_sales = df.groupby("Category")["Sales"].sum()
colors=["red","blue","green"]

category_sales.plot(kind="bar",figsize=(6,4),color=colors)
plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()

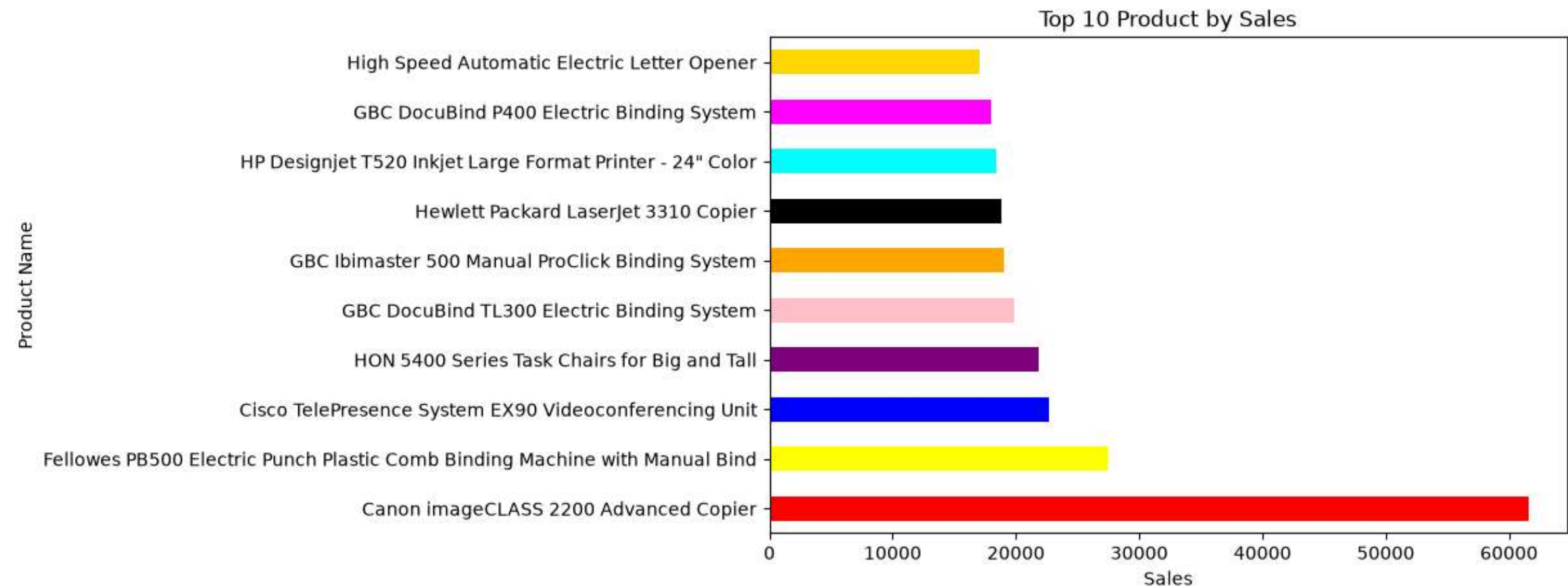


<h2 id="INSIGHT">INSIGHT</h2>

In []: The Technology category generated the highest sales , followed by furniture.office supplies recorded the lowest sales among the three categories.

<h2 id="HORIZONTAL-BARCHART">HORIZONTAL BARCHART</h2><h2 id="**-TOP-10-PRODUCT-BY-SALES**">** TOP 10 PRODUCT BY SALES**</h2>

```
In [8]: top_product = df.groupby("Product Name")["Sales"].sum().sort_values(ascending=False).head(10)
colors=["red","yellow","blue","purple","pink","orange","black","cyan","magenta","gold"]
top_product.plot(kind="barh",figsize=(8,5),color=colors)
plt.title("Top 10 Product by Sales")
plt.xlabel("Sales")
plt.ylabel("Product Name")
plt.show()
```

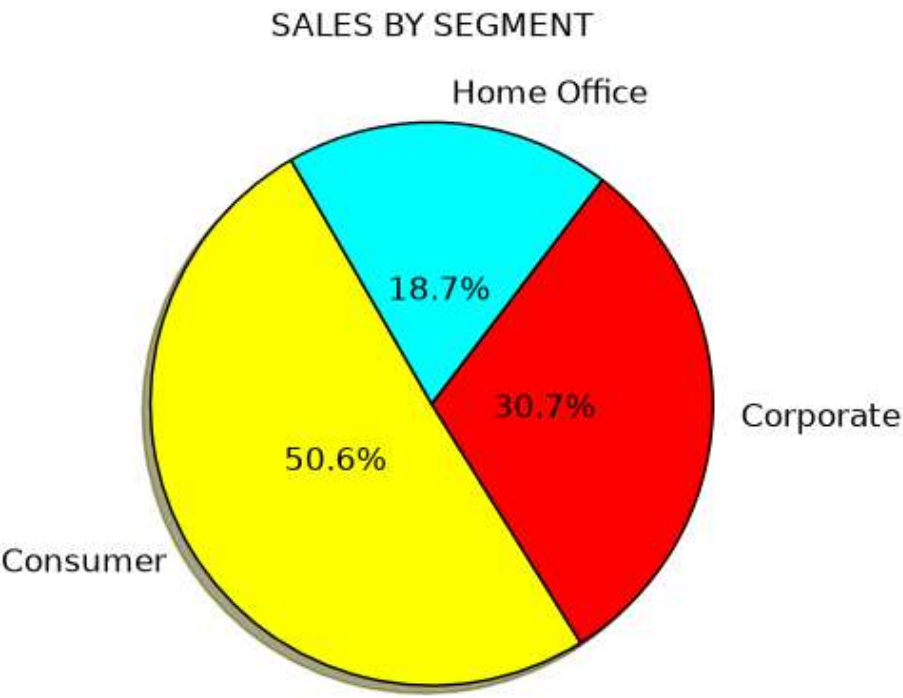


<h2 id="INSIGHT">INSIGHT</h2>

```
In [ ]: Canon image class 2200 advanved copier was highest - selling product ,indicating high customer demand.
```

<h2 id="PIE-CHART">PIE CHART</h2><h2 id="SALES-BY-SEGMENT">SALES BY SEGMENT</h2>

```
In [52]: segment_sales = df.groupby("Segment")["Sales"].sum()
colors = ["yellow","red","cyan"]
segment_sales.plot(kind="pie",autopct="%1.1f%%",labels=segment_sales.index,colors=colors,shadow=True,radius=1,startangle=120,textprops={"fontsize":12},
,wedgeprops={"linewidth":1,"edgecolor":"black"},pctdistance=0.4,labeldistance=1.1,rotatelabels=False)
plt.title("SALES BY SEGMENT")
plt.ylabel("")
plt.show()
```



<h2 id="INSIGHT">**INSIGHT**</h2>

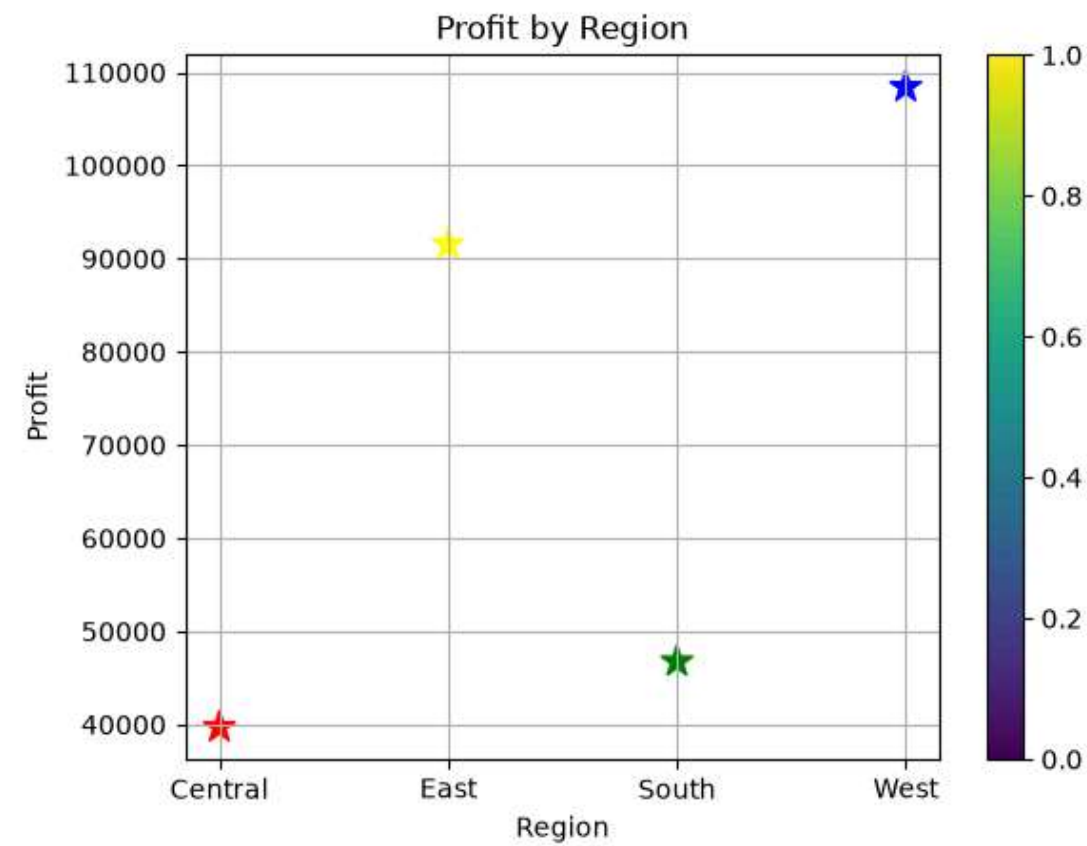
In []: The consumer segment contributed the highest share of sales (about 50.6%), followed by corporate (30.7%) and home office(18.7%).

<h2 id="SCATTER-CHART">**SCATTER CHART**</h2> <h2 id="PROFIT-BY-REGION">**PROFIT BY REGION**</h2>

```
In [9]: region_profit =df.groupby("Region")["Profit"].sum().reset_index()
        colors =["red","yellow","green","blue"]

        plt.scatter(region_profit["Region"],region_profit["Profit"],cmap="virdis",s=150,c=colors,marker="*"),
        plt.colorbar()
        plt.title("Profit by Region")
        plt.xlabel("Region")
        plt.ylabel("Profit")
        plt.grid()
        plt.show()
```

C:\Users\HP\AppData\Local\Temp\ipykernel_4368\2244516707.py:4: UserWarning: No data for colormapping provided via 'c'. Parameters 'cmap' will be ignored
plt.scatter(region_profit["Region"],region_profit["Profit"],cmap="virdis",s=150,c=colors,marker="*"),

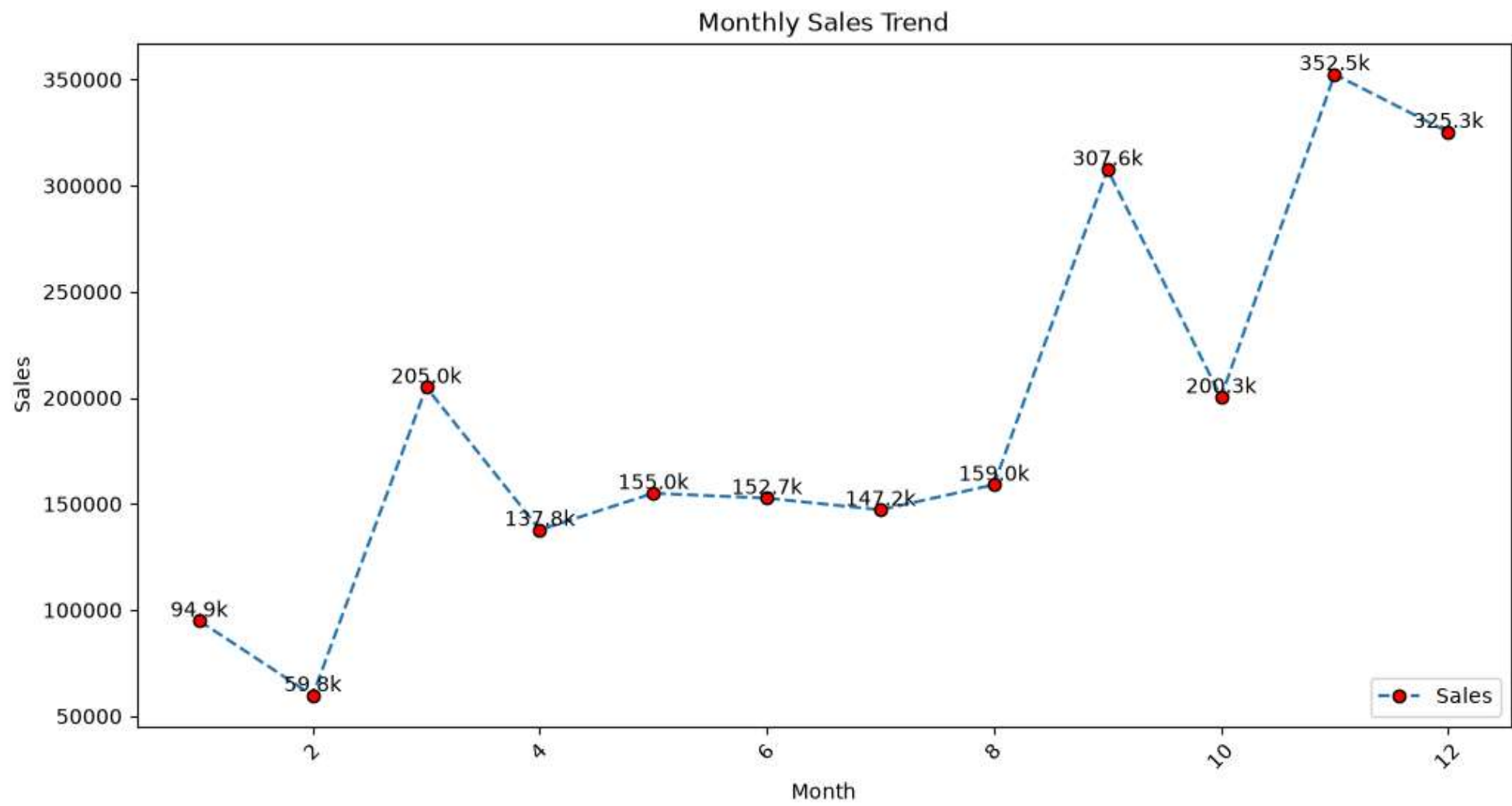


<h2 id="INSIGHT">INSIGHT</h2>

In []: The west region generated the highest profit, while the south region recorded the lowest profit. the company can focus on improving profitability in lower - performing regions.

<h2 id="MONTHLY-SALES-TREND">MONTHLY SALES TREND</h2><h2 id="LINE-CHART">LINE CHART</h2>

```
In [16]: df["Order Date"] = pd.to_datetime(df["Order Date"])
monthly_sales = df.groupby(df["Order Date"].dt.month)["Sales"].sum()
plt.figure(figsize=(12, 6))
monthly_sales.plot(kind="line",marker="o",linestyle="--",markerfacecolor="red",markeredgcolor="black")
for x, y in zip(monthly_sales.index,monthly_sales.values):
    plt.annotate(f"{y/1000:.1f}k", (x,y),xytext=(0,2),textcoords="offset points",ha="center")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.legend(loc="lower right")
plt.show()
```



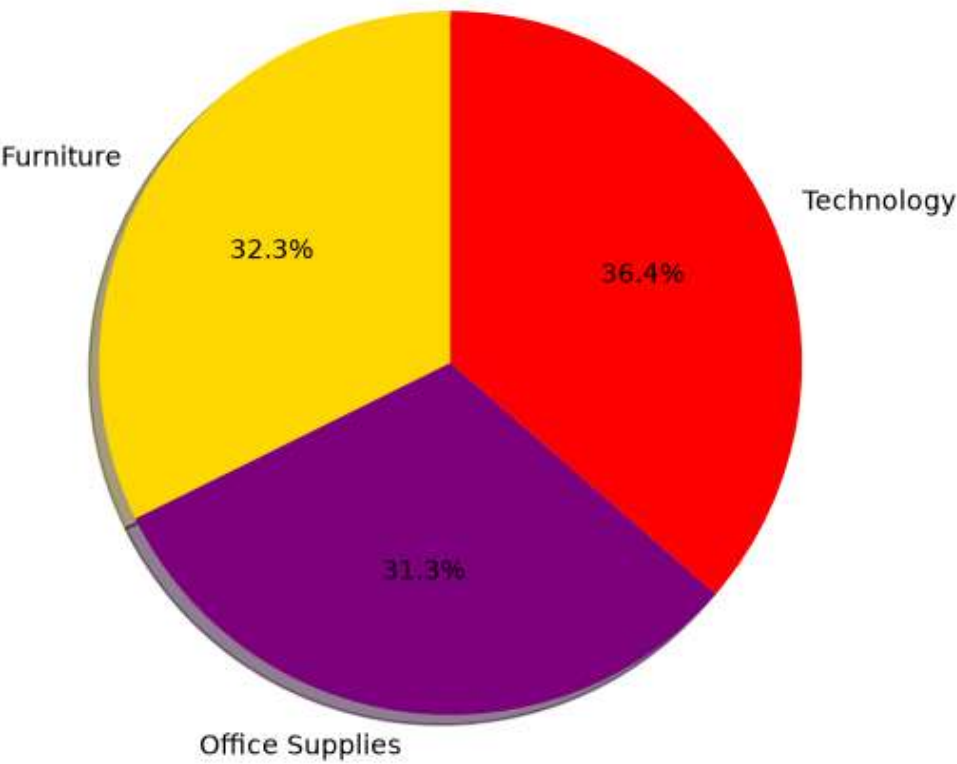
<h2 id="INSIGHT">INSIGHT</h2>

In []: Monthly sales show an overall upward trend with some fluctuations. After a few moderate months, sales increased significantly in the later months, reaching a peak of approximately 352.5K, indicating stronger business performance towards the end of the period.

<h2 id="PIE-CHART">PIE CHART</h2><h2 id="SALES-BY-CATEGORY">SALES BY CATEGORY</h2>

In [4]: category_sales=df.groupby("Category")["Sales"].sum()
plt.figure(figsize=(6,6))
colors=["gold","purple","red"]
plt.pie(category_sales,labels=category_sales.index,autopct="%1.1f%%",startangle=90,colors=colors,shadow=True)
plt.title("Sales by category")
plt.ylabel("")
plt.show()

Sales by category



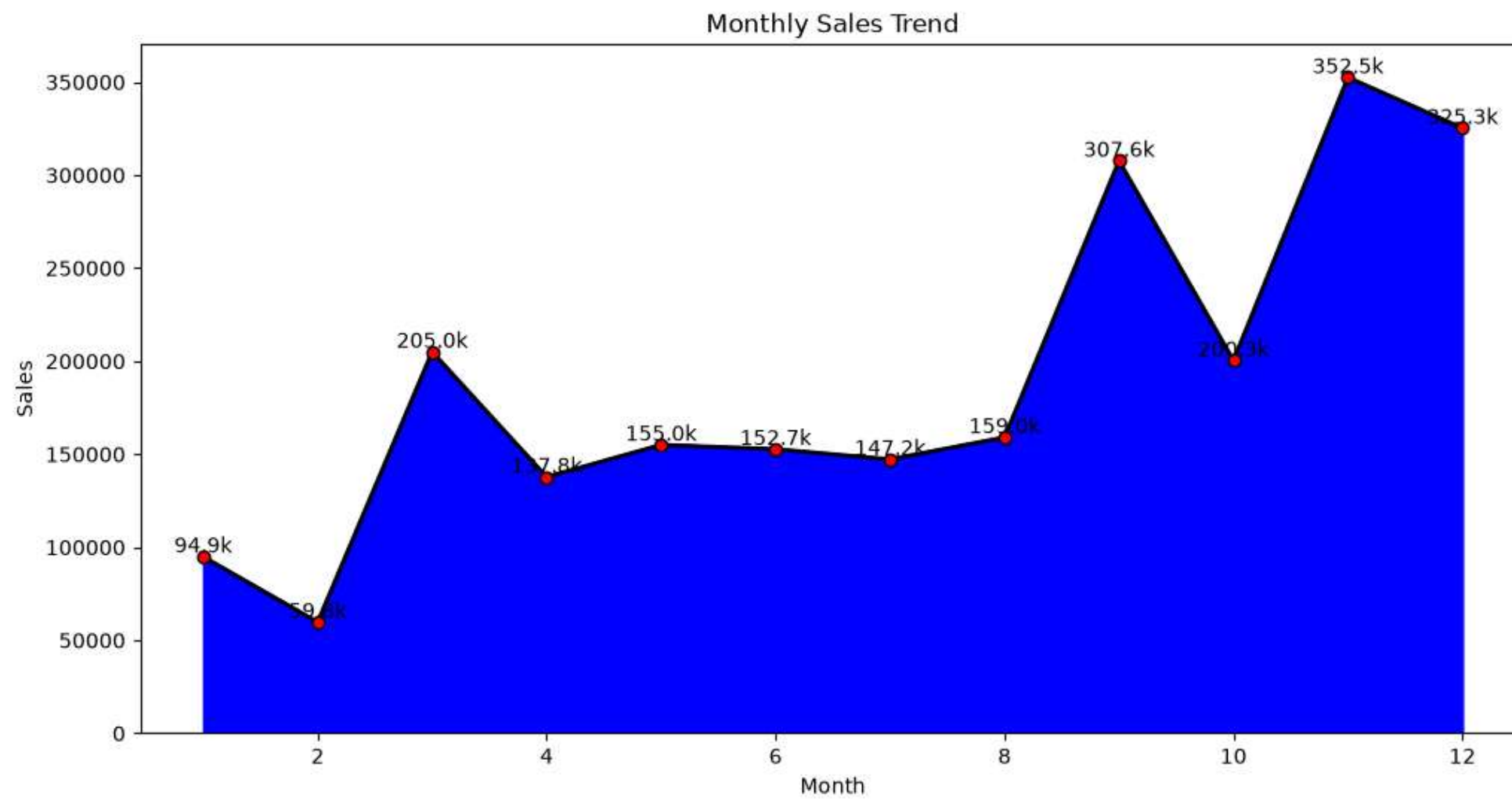
<h2 id="INSIGHT">INSIGHT</h2>

In []: Technology contributed the highest share of total sales (36.4%), while Office Supplies had the lowest share (31.3%).

<h2 id="AREA-CHART">AREA CHART</h2> <h2 id="MONTHLY-SALES-TREND">MONTHLY SALES TREND</h2>

```
In [18]: df["Order Date"] = pd.to_datetime(df["Order Date"])
monthly_sales = df.groupby(df["Order Date"].dt.month)["Sales"].sum()
monthly_sales.plot(kind="area",figsize=(12,6),color="b")
plt.plot(monthly_sales.index,monthly_sales.values,color="black",marker="o",linewidth=2,mfc="r")
for x, y in zip(monthly_sales.index,monthly_sales.values):
    plt.annotate(f"{y/1000:.1f}k",(x,y),xytext=(0,2),textcoords="offset points",ha="center")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



<h2 id="INSIGHT">INSIGHT</h2>

In []: Monthly sales showed an overall upward trend with some fluctuations. Sales reached the highest value (352.5K) in the later months, indicating improved business performance.

<h2 id="BUSINESS-INSIGHT">BUSINESS INSIGHT</h2>

In []: Technology is the highest-selling category, showing strong customer demand. Consumer segment contribute the highest share of sale(50.6%). Sales show strong monthly fluctuation, with the highest sales reaching around \$352.k. The business generated approximately \$286.4k total profit. The comapny should focus on high -performance products, consumer, customer, and better sales and inventory planning.

<h2 id="CONCLUSION">CONCLUSION</h2>

In []: This project successfully analysed the superstore sales dataset using python, pandas and matplotlib. the annalysed identified top - perfoming categories, customer segments ,regions, and products. the insight obtained from the data can support better business decisions and improve overall sales and profitability planning, ultimately improving overall business growth.